

A COUNTEREXAMPLE AT $N = 6$ TO MILIĆEVIĆ'S BLOCKING-POINT CONJECTURE

ABSTRACT. Milićević conjectured that if n points in general position are blocked by n additional points, then all $2n$ points lie on a cubic curve. We give an explicit configuration defined over $\mathbb{Q}(\sqrt{5})$ with $n = 6$ that satisfies the blocking hypothesis but lies on no cubic. Together with the theorem of Bhore and Swanepoel for $n = 5$, this shows that $n = 6$ is the least possible counterexample.

1. THE COUNTEREXAMPLE

Let B be a set of n points in the real affine plane, no three collinear, and let R be a disjoint set of n points such that every line spanned by two points of B contains a point of R . Milićević conjectured that $B \cup R$ lies on a cubic curve [3, Conjecture 5]. Keller and Pinchasi proved the conjecture under the additional requirement that every blue pair has a blocker outside the segment joining it [2, Conjecture 1.2 and Theorem 1.3]. Bhore and Swanepoel proved the case $n = 5$ [1, Theorem 2.9]. For $n \leq 4$, the conclusion follows from the ten-dimensionality of the space of homogeneous plane cubics. Thus the following theorem treats the first possible value of n . Throughout, a cubic is the zero set in $\mathbb{P}^2$ of a nonzero homogeneous polynomial of degree three; reducible and nonreduced cubics are included.

Theorem 1. *There exist disjoint six-point sets B and R in the real affine plane such that no three points of B are collinear and every line spanned by two points of B contains a point of R , but $B \cup R$ is contained in no cubic curve.*

Proof. Set

$$p = \frac{3 - \sqrt{5}}{2}.$$

Then $p^2 = 3p - 1$ and $0 < p < \frac{1}{2}$. In $\mathbb{P}^2(\mathbb{R})$, let

$$\begin{aligned} b_1 &= (1 : 0 : 0), & b_2 &= (0 : 1 : 0), & b_3 &= (0 : 0 : 1), \\ b_4 &= (1 : 1 : 1), & b_5 &= (p : 1 - p : 1), & b_6 &= (1 - p : p : 1), \end{aligned}$$

and

$$\begin{aligned} r_1 &= (p : 0 : 1), & r_2 &= (1 - p : 1 : 1), & r_3 &= (1 : 1 - p : 1), \\ r_4 &= (0 : p : 1), & r_5 &= (-1 : -1 : 0), & r_6 &= (1 : 1 : 2). \end{aligned}$$

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The three points with $z = 0$ are distinct. After normalization to $z = 1$, the remaining nine points are also distinct because $0 < p < \frac{1}{2}$. Hence all twelve points are distinct. Moreover, $x + y + z$ takes only the nonzero values

$$1, 2, 3, 1 + p, 3 - p, -2, 4$$

on these points. We therefore regard $\mathbb{P}^2 \setminus \{x + y + z = 0\}$ as a common affine chart.

For $i < j < k$, write

$$[ijk] = \det(b_i, b_j, b_k),$$

with the coordinate vectors as columns. Reduction by $p^2 = 3p - 1$ gives the complete list

value of $[ijk]$	triples ijk
1	123, 124, 125, 126, 234
-1	134
p	145, 235
$-p$	136, 246
$1 - p$	146, 236
$-(1 - p)$	135, 245
$1 - 2p$	156, 256, 345, 456
$-(1 - 2p)$	346, 356

All these values are nonzero, so no three points of B are collinear.

Write ij for the line through b_i and b_j . Direct evaluation of $\det(b_i, b_j, r_k)$ gives exactly the following incidences:

red point	blue pairs blocked
r_1	13, 25, 46
r_2	14, 26, 35
r_3	15, 24, 36
r_4	16, 23, 45
r_5	12, 34
r_6	34, 56

Every one of the fifteen blue pairs occurs, so R blocks all lines spanned by B . The stronger hypothesis of Keller and Pinchasi is not satisfied: using the normalized representatives with $x + y + z = 1$, one has

$$r_5 = \frac{b_1 + b_2}{2}, \quad r_6 = \frac{b_5 + b_6}{2},$$

and the incidence table shows that these are the only red points on 12 and 56, respectively.

It remains to exclude a cubic. Suppose that a nonzero homogeneous cubic $F(x, y, z)$ vanishes at all twelve points. The four distinct points

$$b_3, \quad b_4, \quad r_5, \quad r_6$$

lie on the line $x - y = 0$. The restriction of F to this line is a binary cubic with four distinct zeros, hence is identically zero. Therefore

$$F = (x - y)Q, \quad Q = Ax^2 + By^2 + Cz^2 + Dxy + Exz + Gyz.$$

The points $b_1, b_2, r_1, r_4, b_5, r_2$ lie off $x - y = 0$, so Q vanishes at each of them. Evaluation at b_1 and b_2 gives $A = B = 0$, while evaluation at r_1 and r_4 gives

$$C + pE = 0, \quad C + pG = 0.$$

If $C = 0$, then $E = G = 0$, and evaluation at b_5 gives $Dp(1 - p) = 0$, hence $D = 0$, contradicting $F \neq 0$. Thus $C \neq 0$. After scaling, take $C = p^2$. Then $E = G = -p$, and evaluation at b_5 gives

$$0 = Q(b_5) = p(1 - p)(D - 1),$$

so $D = 1$. Consequently,

$$Q = xy - pxz - pyz + p^2z^2 = (x - pz)(y - pz).$$

At $r_2 = (1 - p : 1 : 1)$, however,

$$F(r_2) = (-p)(1 - 2p)(1 - p) \neq 0,$$

a contradiction. Hence no cubic contains $B \cup R$. □

REFERENCES

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